

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY  
BELAGAVI - 590018**



**PROJECT REPORT ON**  
**“RFID BASED PETROLEUM PUMP AUTOMATION SYSTEM”**  
Submitted in partial fulfilment of the requirement for the award of the degree of  
**BACHELOR OF ENGINEERING**  
**IN**  
**ELECTRONICS AND COMMUNICATION ENGINEERING**

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**2025-26**



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# GOVERNMENT ENGINEERING COLLEGE

## Department of Electronics and Communication Engineering



### CERTIFICATE

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As a part of 7<sup>th</sup> semester course in the degree **Bachelor of Engineering in Electronics and Communication Engineering** in **Government Engineering College, Hassan** of **Visvesvaraya Technological University, Jnana Sangma, Belagavi-590018** during the year **2025-26**.

It is certified that all the corrections or suggestions indicated for the internal assessment has been approved as it satisfies the academic requirements with respect to project work prescribed for the said degree.

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2) .....

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## ABSTRACT

The RFID-based petrol pump automation system aims to transform traditional fuel dispensing into an efficient, secure, and fully automated process. Conventional petrol pump operations often involve manual verification, billing delays, human errors, and chances of fuel misuse. To overcome these challenges, this project integrates Radio Frequency Identification (RFID) technology with microcontroller-based control, sensor monitoring, and a centralized database. In this system, each vehicle is assigned a unique RFID tag linked to the owner's account or prepaid balance. When the vehicle reaches the petrol station, the RFID reader automatically scans the tag and authenticates the user without the need for manual intervention. Once authentication is verified, the system initiates fuel dispensing according to the user's account balance or predefined fuel limits. The microcontroller continuously monitors the dispensing process, and the system automatically calculates the fuel quantity, amount, and updates the user's balance in real time. The database records every transaction, improving transparency and reducing chances of fraud or unauthorized fuel access. This automation enhances accuracy, speeds up service time, reduces manpower requirements, and increases overall operational safety. The project highlights the practical application of RFID, embedded systems, and automation technologies to create a smart, user-friendly, and reliable petrol pump management solution. It showcases how modern IoT and sensor integration can contribute to developing efficient fuel distribution systems for future smart cities.

**Keywords:** RFID, Petrol Pump Automation, Arduino UNO, Embedded Systems, Automated Fuel Dispensing, RFID Reader, Microcontroller, Cashless Transaction, Fuel Management System.

## ACKNOWLEDGEMENT

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# LIST OF CONTENTS

|                                |           |
|--------------------------------|-----------|
| <b>ABSTRACT</b>                | <b>i</b>  |
| <b>ACKNOWLEDGEMENT</b>         | <b>ii</b> |
| <b>CHAPTER 1: INTRODUCTION</b> | <b>1</b>  |
| <b>CHAPTER 2:</b>              | <b>2</b>  |
| 2.1 LITERATURE SURVEY          | 2         |
| 2.2 PROBLEM STATEMENT          | 5         |
| 2.3 OBJECTIVES                 | 5         |
| <b>CHAPTER 3: METHODOLOGY</b>  | <b>7</b>  |
| 3.1 PRINCIPLE                  | 7         |
| 3.2 COMPONENTS                 | 7         |
| 3.3 BLOCK DIAGRAM              | 14        |
| 3.4 FLOW CHART                 | 17        |
| 3.5 PROPOSED ARCHITECTURE      | 18        |
| <b>CHAPTER 4: RESULTS</b>      | <b>20</b> |
| 4.1 RESULT                     | 20        |
| 4.2 ADVANTAGES                 | 21        |
| 4.3 DISADVANTAGES              | 22        |
| 4.4 APPLICATIONS               | 23        |
| <b>CONCLUSION</b>              | <b>24</b> |
| <b>FUTURE SCOPE</b>            | <b>24</b> |
| <b>REFERENCES</b>              | <b>26</b> |

## LIST OF FIGURES

|                                                                           |    |
|---------------------------------------------------------------------------|----|
| Figure 3.1: Arduino UNO                                                   | 7  |
| Figure 3.2: EM-18 RFID Reader                                             | 8  |
| Figure 3.3: Keypad                                                        | 9  |
| Figure 3.4: LCD Display                                                   | 10 |
| Figure 3.5: Relay Module                                                  | 10 |
| Figure 3.6: Buzzer                                                        | 11 |
| Figure 3.7: Regulated Power Supply                                        | 12 |
| Figure 3.8: Fueling Pump                                                  | 12 |
| Figure 3.9: Jumper Wires                                                  | 13 |
| Figure 3.10: Arduino IDE                                                  | 14 |
| Figure 3.11: Block Diagram of RFID based Petroleum Pump Automation System | 15 |
| Figure 3.12: Flowchart of RFID based Petroleum Pump Automation System     | 19 |
| Figure 3.13: Architecture                                                 | 20 |

## CHAPTER 1

# INTRODUCTION

Radio Frequency Identification (RFID) is an advanced automatic identification technology that uses radio waves to recognize and track objects without requiring direct contact or line-of-sight communication. In recent years, RFID has been widely implemented in various industries including transportation, supply chain management, banking, and automation. Its ability to provide fast, secure, and contactless identification makes it highly suitable for modern automated systems.

Petrol pumps traditionally rely on manual procedures such as customer verification, billing, payment processing, and fuel dispensing. These manual operations can lead to inefficiencies, human errors, fuel theft, and delays in service. Additionally, there is a growing need for transparency, accurate fuel measurement, and secure transaction handling in fuel stations. To address these challenges, automation using RFID technology offers a reliable and efficient solution.

An RFID-based petrol pump automation system automates the entire fueling process. Each authorized vehicle is equipped with an RFID tag that contains a unique identification code fuel balance. The pump remains active only for the required amount of fuel, ensuring accuracy and preventing unauthorized usage.

The system maintains digital transaction records, enhances operational efficiency, and reduces dependency on manual intervention. It also increases security by ensuring that only authenticated and registered users can access fuel. Furthermore, the automated process helps reduce waiting time, improves resource management, and supports digital payment integration. Overall, the RFID-based petrol pump automation system represents an innovative technological advancement aimed at modernizing fuel distribution processes. By integrating RFID technology with embedded systems, it provides a secure, efficient, and transparent method of managing fuel dispensing, making it a valuable solution for both commercial fuel stations and large-scale vehicle fleets



## CHAPTER-2

### 2.1 LITERATURE SURVEY

A. Kumar and S. Rao [1] Presented the design of an RFID-enabled fuel dispensing automation system. Their research focused on integrating RFID readers with embedded controllers to automatically authenticate users and control the fuel dispensing process. The system architecture reduced manual intervention and improved dispensing accuracy, forming a strong base for RFID-based embedded automation systems.

M. Patel and R. Shah [2] Proposed a secure payment system using RFID cards for automated petrol pumps. Their work emphasized secure transaction handling, encryption techniques, and authentication mechanisms to prevent financial fraud. The research mainly addressed cashless payment automation rather than direct control of the pump hardware.

T. Singh and V. Pillai [3] Developed an automated fuel station system using microcontrollers and RFID technology. Their system verified users through RFID cards and enabled automatic fuel dispensing. The study demonstrated improved operational efficiency, reduced human error, and faster service at fuel stations.

S. R. Bhat and K. Ramesh [4] Introduced an IoT-based monitoring and control system for fuel pumps. Their research enabled real-time monitoring of fuel levels, pump operation, and transaction data through internet connectivity. Although the system improved remote supervision and fault detection, it required continuous network availability and higher implementation cost.

P. Khanna and M. Verma [5] Focused on RFID authentication for secure petrol distribution. Their research addressed security challenges such as unauthorized access and card duplication. The proposed system ensured that fuel dispensing occurs only after successful RFID verification, thereby enhancing system security.

D. Roy and A. Banerjee [6] Proposed a real-time fuel management system using RFID and

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GSM modules. Their system provided remote communication and transaction updates through SMS alerts. This research highlighted the role of wireless communication in improving fuel station monitoring and management.

K.Srinivas and P. K. Das [7] Designed a microcontroller-based automated fuel dispenser using RFID. Their system used relay-based control mechanisms to manage fuel flow based on RFID authentication. The research demonstrated a cost-effective and reliable solution suitable for small-scale fuel automation systems.

L.Mathew and G. George [8] Proposed an RFID-based vehicle identification system for fuel stations. Their research linked RFID tags directly to vehicles, ensuring that fuel is dispensed only to registered vehicles. This approach improved accountability and prevented misuse of fuel resources.

N. Ahmed and S. Ali [9] Developed an RFID-based automation system for oil and gas dispensing. Their research focused on large-scale fuel dispensing systems, emphasizing improved safety, monitoring, and operational accuracy through RFID technology.

P. S. Kumar and A. Joseph [10] Proposed a contactless payment system using RFID for fuel automation. Their research aimed at eliminating cash transactions by linking RFID cards with prepaid or postpaid accounts. The system enhanced user convenience and reduced transaction time. However, their work focused more on payment processing and backend data handling. In contrast, our project concentrates on physical fuel dispensing control using a relay and motor, making it suitable for basic automation applications.

V. Reddy and H. R. Kulkarni [11] Designed a petrol pump automation system using Arduino and RFID technology. Their system automatically authorized users through RFID cards and controlled fuel dispensing using a microcontroller. This research directly influenced our project, as we adopted a similar approach using an Arduino board, RFID reader, and relay module to control the pump motor. Their work proved the feasibility of RFID-based automation in real-time fuel dispensing.

S. Prakash and R. Sen [12] Developed an enhanced fuel delivery monitoring system using RFID integrated with IoT technology. Their system enabled real-time data transmission, remote monitoring, and cloud storage of fuel transactions. This approach increased transparency and accountability in fuel distribution. Although our project does not include IoT features, their research highlighted the importance of monitoring and data management, which can be considered as a future enhancement of our system.

A. Chatterjee and D. Gupta [13] Focused on RFID-based fraud prevention techniques in automated fuel stations. Their research addressed challenges such as unauthorized fueling, duplicate RFID cards, and tampering. They proposed secure authentication methods to ensure that fuel is dispensed only to valid users. The security concepts discussed in this work influenced the authentication mechanism used in our project, ensuring that the pump activates only after successful RFID verification.

T. Fernandes and L. Costa [14] Proposed a vehicle-linked RFID fueling system, where RFID tags are permanently associated with specific vehicles rather than users. This method improved security by preventing misuse of RFID cards across multiple vehicles. Their research demonstrated better accountability in fuel usage. While our project uses userbased RFID cards, the concept of controlled access aligns with our system's objective of authorized fuel dispensing.

R. Mishra and A. Yadav [15] Presented a smart fuel station system using microcontrollers and RFID. Their work focused on automating fuel dispensing, maintaining transaction records, and reducing human involvement. The research supported the use of embedded systems for automation, validating our choice of Arduino as the controller for managing RFID input and pump operation.

H. Singh and P. K. Nair [16] Studied efficiency improvement in fuel stations using RFIDbased automation. Their research concluded that automation significantly reduces waiting time, operational cost, and manual workload. This study supports the motivation behind our project, which aims to provide faster and more accurate fuel dispensing with minimal human intervention.

S. Latha and D. Kumar [17] Developed a secure fuel filling system using smart cards and RFID technology. Their work emphasized system reliability, secure authentication, and user data protection. The concepts of secure card-based access from their research were incorporated into our project to ensure that only authorized RFID cards can activate the pump.

## **2.2 PROBLEM STATEMENT**

Traditional petrol pump operations rely heavily on manual supervision for vehicle identification, fuel dispensing, billing, and transaction monitoring. This manual process often leads to challenges such as long waiting times, human errors in billing, fuel theft, unauthorized access, and inefficient record maintenance. Customers may face delays due to slow processing, while petrol pump owners struggle to maintain accurate data regarding fuel usage, customer identity, and financial transactions.

In the absence of an automated system, there is no reliable mechanism to ensure prepaid fueling, track customer accounts, verify authentication, or prevent fuel misuse. This creates operational inefficiencies and reduces overall transparency in the fuel distribution process. Hence, there is a need for an automated, secure, and user-friendly solution that can identify customers using RFID technology, validate account balance, allow prepaid fueling, automatically dispense fuel according to the amount entered, and generate accurate digital records eliminating human dependency and improving operational reliability.

The proposed RFID-Based Petrol Pump Automation System aims to solve these issues by providing seamless authentication, secure transactions, automatic fuel control, and realtime data management.

## **2.3 OBJECTIVES**

The main objective of the RFID-Based Petrol Pump Automation System is to introduce an efficient, secure, and fully automated method for fuel dispensing by reducing human intervention and improving system reliability. This project aims to modernize conventional petrol pump operations using RFID technology and automated control mechanisms.

1. To automate the customer identification process using RFID cards for secure and accurate authentication.
2. To eliminate manual errors in fuel dispensing and billing through automated digital

control.

3. To provide a prepaid fueling system where users can refuel only based on available balance.
4. To ensure accurate measurement of fuel and prevent manipulation or theft during dispensing.
5. To record all transaction data digitally for future reference, auditing, and transparency.
6. To maintain a centralized database of users, fuel usage, account balance, and transaction history
7. To integrate user-friendly interfaces for both customers and petrol pump operators.
8. To support future upgrades such as IoT monitoring, cloud-based data storage, and mobile app connectivity.
9. To reduce operational costs by automating tasks and reducing manpower dependency.
10. To increase transparency in billing and fuel distribution by providing an automation

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## CHAPTER-3

# METHODOLOGY

### 3.1 PRINCIPLE

The RFID-based petrol pump automation system works on the principle of automatic identification and secure fuel dispensing using RFID technology.

Once the ID is validated, the microcontroller prompts the user to enter a password and the required fuel amount through the keypad. After processing this information, the controller activates the relay, which switches ON the pump (motor) for the exact duration needed to dispense the entered fuel amount.

During the entire process, the LCD displays system messages, and the buzzer alerts the user for any errors or completion. When the required amount is dispensed, the microcontroller turns OFF the pump and updates the transaction. Thus, the system ensures accurate, secure, and automated fuel delivery without manual handling.

### 3.2 COMPONENTS

#### Hardware Components :

##### 1. Arduino UNO



**Fig 3.1: Arduino UNO**

The Arduino checks whether the ID is valid and then prompts the user (through the LCD) to enter a password and the required fuel amount using the keypad.

After receiving the inputs, the Arduino processes the data and calculates how long the pump (DC motor) needs to run to dispense the exact fuel amount. It then sends a control signal to the relay module to switch the pump ON. The Arduino continuously monitors

the dispensing time and turns the pump OFF automatically once the required amount is delivered.

Throughout the process, the Arduino updates the LCD with messages such as authentication status, instructions, dispensing progress, and completion. It also triggers the buzzer in case of wrong entries or alerts.

## 2. EM-18 RFID Reader



**Fig 3.2: EM-18 RFID Reader**

The EM-18 RFID Reader module is one of the most important components in the RFID based petrol pump automation system. It is responsible for identifying and authenticating users or vehicles through RFID cards or tags. The main function of the EM-18 RFID reader is to provide secure and contactless identification.

In the petrol pump automation system, the EM-18 reader is used to authenticate users before allowing fuel dispensing. When a valid card is detected, the microcontroller compares the UID with the stored database. If the card matches, the system allows the user to proceed by entering fuel quantity through the keypad. If the card is invalid or unauthorized, the EM-18 reader helps the system block access and alert the user, ensuring security and preventing misuse. balance, or usage limit, depending on the system design. It helps in maintaining accuracy and transparency during fuel distribution. By using RFID cards, the system eliminates the need for manual verification and provides a faster, more reliable method of identifying users.

Overall, the function of the RFID card in the system is to provide secure user authentication, enable automated fuel authorization, and maintain controlled access to the petrol pump, enhancing both security and efficiency of fuel management.

### 3. Keypad



**Fig 3.3 : Keypad**

Keypad enables the operator or customer to manually enter data such as fuel quantity, password, or system commands. The keypad provides an easy and user-friendly way to interact with the microcontroller and helps in controlling the fuel dispensing process accurately. Apart from quantity entry, the keypad can also be used for password protection and system security. An administrator password may be required to access system settings, reset the device, or perform manual overrides. This helps prevent unauthorized usage and ensures safe operation.

### 4. LCD Display



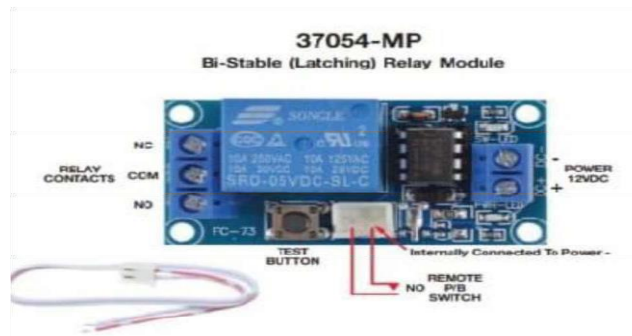
**Fig 3.4: LCD display**

The I2C module with LCD is used as a display unit that provides real-time information to the user and operator. The LCD display (usually 16×2 or 20×4) shows instructions, messages, fuel quantity, card status, and system notifications. The addition of the I2C interface module simplifies the connection between the LCD and the microcontroller by reducing the number of pins required. In this project, the I2C LCD displays important system information at every stage of fuel quantity or checks available balance in prepaid systems. During fuel dispensing, the screen shows the status of pump operation, the



running process, and once completed, it displays “fuel dispensed successfully”

## 5. Relay Module



**Fig 3.5: Relay Module**

The relay plays a crucial role as an electrically operated switch that controls the operation of the fuel pump. The microcontroller used in the system works with low voltage signals, typically 3.3V or 5V, which are not sufficient to directly drive highpower devices like fuel pumps or solenoid valves. Therefore, a relay module is used to interface the low-power. When a valid RFID card is scanned and the user enters the fuel quantity, the microcontroller processes this information and sends a signal to activate the relay. Once activated, the relay closes the normally open (NO) contact, thereby powering the pump and allowing fuel to flow. After the required quantity is dispensed, the relay is de-energized, which opens the circuit and stops the pump.

## 6. Buzzer



**Fig 3.6: Buzzer**

The buzzer is used as an audio indication device that provides sound alerts during various stages of the operation. It plays an important role in enhancing the user interface and ensuring proper system feedback. The buzzer is typically a small piezoelectric or magnetic buzzer that operates on low voltage (3.3V or 5V) and is controlled by the microcontroller. The primary function of the buzzer is to give audio confirmation signals whenever a specific action occurs within the system. For example, when an RFID tag is detected

successfully, the buzzer emits a short beep to notify the user that the card has been read.

## 7. Regulated Power Supply



**Fig 3.7: Regulated Power Supply**

The 12V regulated power supply is a crucial part of the RFID-based petrol pump automation system, providing stable and reliable electrical power to all components that require 12 volts for proper operation. Many modules in the system, such as the relay module, solenoid valve, and pump motor, require a steady 12V DC supply. The regulated power supply ensures that these components receive a constant voltage without fluctuations, which is important for both performance and safety.

The power supply unit typically consists of a step-down transformer, bridge rectifier, filter capacitor, and a voltage regulator circuit. The AC mains voltage (230V) is first reduced to a lower AC voltage, then rectified to DC, filtered to remove ripples, and finally regulated to a stable 12V DC output. This regulated output protects the system from voltage spikes, overloads, and irregular supply conditions that could damage the electronic components.

## 8. Fueling Pump



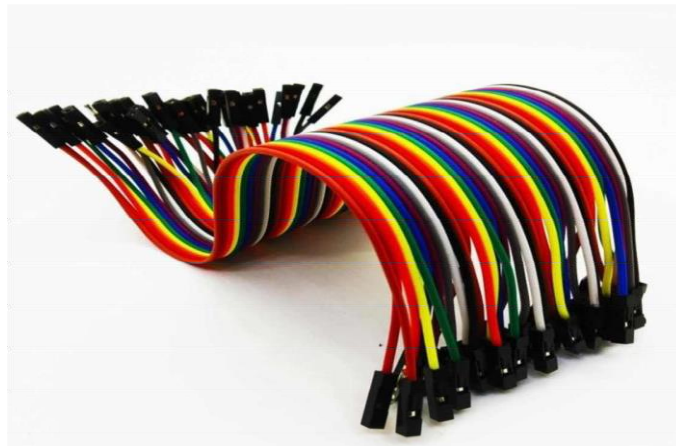
**Fig 3.8: Fueling Pump**

The DC motor pump is a key electromechanical component responsible for enabling the fuel dispensing operation in the RFID-based petrol pump automation system. Once the authorized user is identified through the RFID module, the microcontroller activates the DC motor pump to initiate the transfer of fuel from the storage tank to the outlet nozzle.

The pump converts electrical energy supplied to the DC motor into mechanical energy,

state and operating duration, the microcontroller regulates the amount of fuel delivered. This allows automated, accurate, and tamper-quantity or available balance. The motor pump therefore functions as the final actuation stage of the system, ensuring smooth, controlled, and automated fuel delivery. Additionally, once the preset quantity is achieved, the microcontroller stops the motor, preventing overflow and ensuring safety.

## 9. Jumper Wires



**Fig 3.9: Jumper Wires**

Jumper wires are essential electrical conductors used in electronic prototyping to establish temporary connections between different components in a circuit. They provide a simple and flexible means of linking modules, sensors, and microcontrollers without the need for soldering. Jumper wires are available in different types such as male male, male female, and female female, allowing compatibility with both breadboards and header pins.

In the RFID-based petrol pump automation system, jumper wires play a crucial role in interconnecting the RFID reader, keypad, LCD display, relay module, and the Arduino microcontroller. They carry power, ground, and signal lines between these modules, ensuring smooth communication and reliable operation of the entire system. By enabling quick modifications and easy assembly, jumper wires support efficient prototyping and testing of the project.

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**SOFTWARE:****10.Arduino IDE**

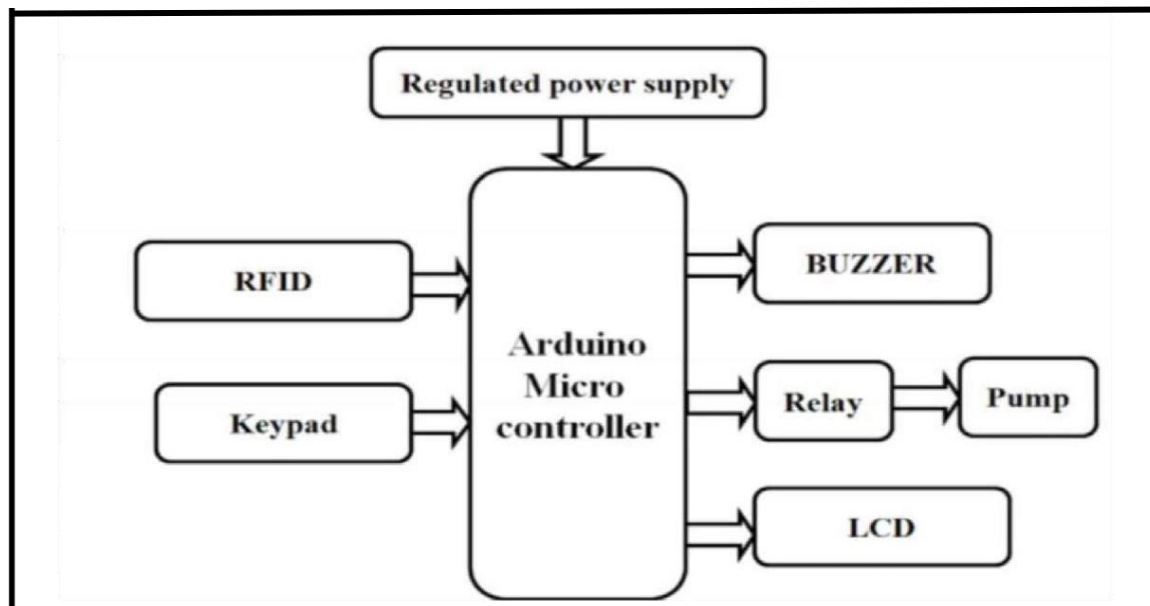
**Fig 3.10: Arduino IDE**

The Arduino Integrated Development Environment (IDE) is the official software platform used for programming and uploading code to Arduino microcontroller boards. It provides a user-friendly interface for writing, editing, compiling, and transferring programs to the Arduino hardware. The Arduino IDE supports the Arduino programming language, which is based on C and C++, and enables rapid development of embedded system applications.

The IDE contains a built-in text editor where the user writes the program, known as a sketch. The software includes essential tools such as Verify/Compile, which checks the program for syntax errors, and Upload, which transfers the compiled machine code to the microcontroller through a USB connection. The Arduino IDE automatically manages the conversion of high level code into a hex file, which the microcontroller executes.

A key feature of the Arduino IDE is the Serial Monitor, which allows real-time communication between the computer and the Arduino board. It is widely used for debugging, displaying sensor values, and monitoring system performance.

### 3.3 BLOCK DIAGRAM



**Fig 3.11: Block Diagram of RFID Based Petroleum Pump Automation System**

The block diagram shows the overall working structure of the RFID-based petrol pump automation system. The Arduino microcontroller is the central unit that connects all the input and output modules. The inputs to the controller come from the RFID reader, which identifies the user, and the keypad, which allows the user to enter the password and fuel amount.

Using these inputs, the Arduino processes the data and sends control signals to the output devices. The relay controls the pump (DC motor) to dispense fuel according to the entered amount. The LCD displays messages and guides the user throughout the process. The buzzer provides audio feedback for events like errors or completion.

All these modules operate through a regulated power supply, ensuring stable power to the entire system. The diagram represents how each block interacts with the controller to complete the fuel dispensing operation securely and automatically.

#### **Working:**

The working of the RFID-based petrol pump automation system is based on the interaction between the RFID reader, keypad, Arduino microcontroller, relay, LCD, buzzer, and fuel pump. The regulated power supply provides stable operating voltage to all components. The step-by-step working is described below:

#### **1. Power Initialization**

The regulated power supply provides a stable 5V/12V output to the Arduino, RFID reader,

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keypad, LCD, buzzer, and relay circuit.

Once powered ON, the microcontroller initializes all connected modules and displays a welcome message on the LCD screen.

## **2. RFID Authentication Process**

- The user presents an RFID card near the RFID reader.
- The RFID reader scans the card and sends the unique ID number to Arduino microcontroller.
- The Arduino compares the received ID with the database stored in its memory.
- If the RFID card is “valid”:
- The LCD displays “Access Granted”
- The buzzer gives a short beep indicating successful authentication.
- The system moves to the next stage (amount entry).
- If the RFID card is invalid:
- The LCD Displays access “Denied”
- The buzzer sounds continuously to indicate a security error.
- The pump remains OFF and the process stops here.

## **3. Amount/Quantity Entry via Keypad**

- After successful RFID verification, the system prompts the user to enter the required fuel amount or quantity using the keypad.
- The keypad sends numerical input data to the Arduino.
- The Arduino verifies the entered amount and checks the available balance linked with the RFID card.
- If sufficient balance is available:
- The system proceeds to start fuel dispensing.
- If balance is insufficient:
- The LCD shows “Low Balance”
- The buzzer alerts the user.
- The process stops without activating the pump.

## **4. Activation of Relay and Fuel Pump**

- Once the amount and balance are validated, the Arduino sends a control signal to the relay module.
- The relay acts as an electronic switch that turns ON the fuel pump.
- Fuel dispensing begins, and the Arduino monitors the pumping duration or amount based on preprogrammed calibration.

**5. Fuel Dispensing and Monitoring**

- The system dispenses fuel only up to the amount entered by the user.
- The microcontroller continuously monitors the pumping process.
- Once the required fuel quantity/amount is delivered.
- The Arduino sends a LOW signal to the relay. The relay switches OFF the pump immediately.

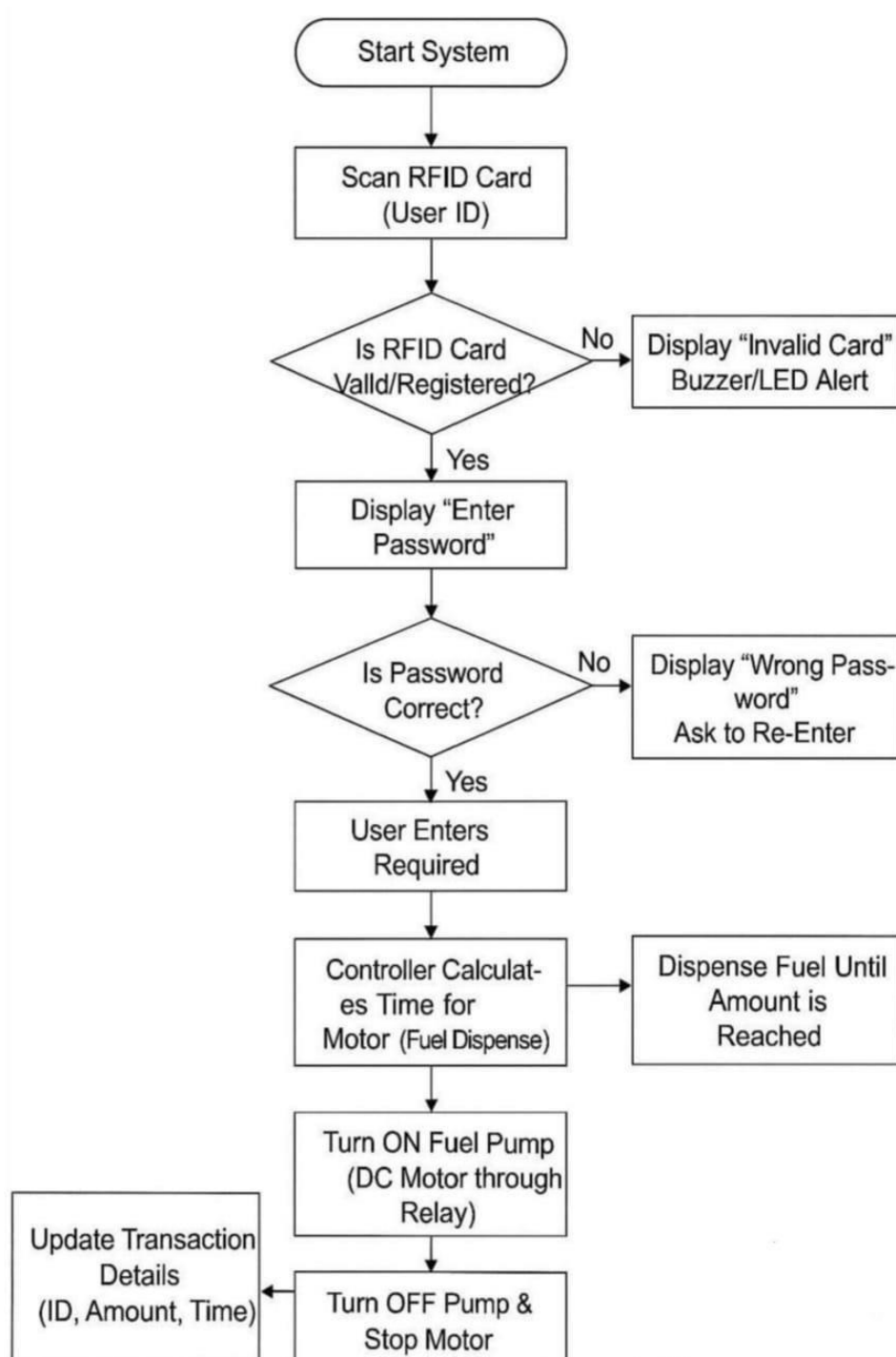
**6. Display of Transaction Details**

After completion, the LCD displays:

- Fuel dispensed
- Amount deducted
- Remaining balance in RFID card
- The transaction data is stored in the Arduino memory (or optional database).

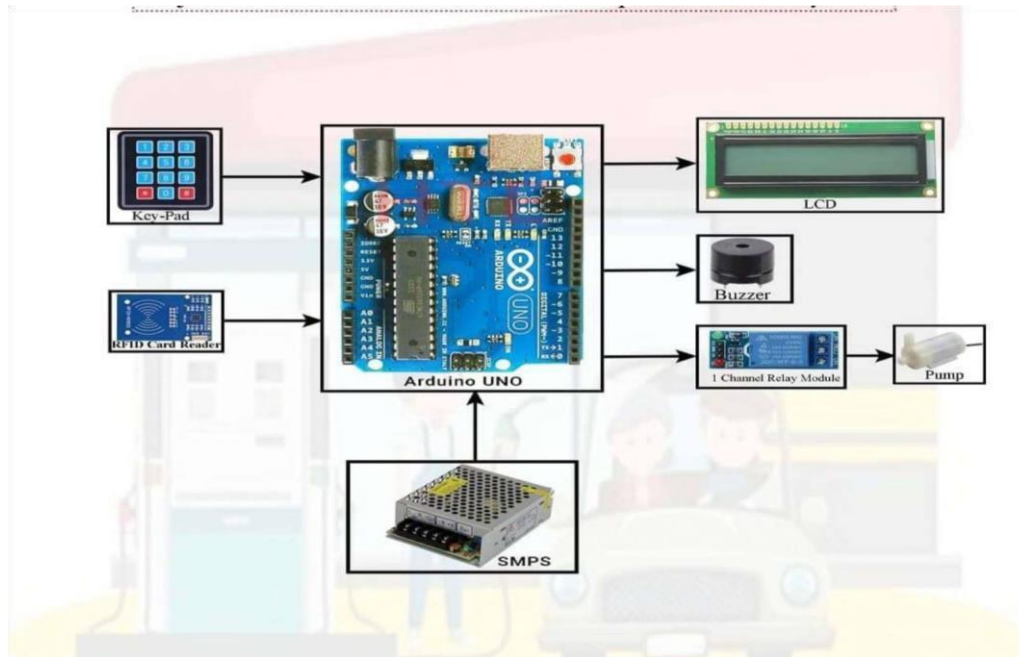
**7. Completion and System Reset**

- The buzzer gives a short beep indicating successful completion of the refueling.
- The system resets back to the initial waiting state, ready for the next user.

**3.4 FLOW CHART:****Fig 3.12: Flow Chart of RFID Based Petroleum Pump Automation System**



### 3.5 PROPOSED ARCHITECTURE



**Fig 3.13: Architecture of Proposed Model**

The architecture of the RFID-based petrol pump automation system is divided into four major functional blocks: User Identification Unit, Central Control Unit, Fuel Dispensing Unit, and Communication & Data Management Unit. These units operate together to provide authenticated access, automated fuel delivery, safety monitoring, and accurate transaction recording.

#### 1. User Identification Unit

This unit is responsible for identifying and authenticating the customer before fuel dispensing begins.

##### Functions

- Extracts the unique identification data from the RFID medium.
- Sends this identification data to the central control system.
- Ensures that only valid, registered, or prepaid users are allowed to proceed.
- This block provides secure access control to the fuel dispensing system.

#### 2. Central Control Unit

This is the decision-making and coordination unit of the system.

##### Functions

- Receives user identification details and verifies authenticity.
- Checks user account status, fuel limit, or authorization conditions.
- Initiates the fuel dispensing process upon successful authentication.
- Continuously monitors real-time fuel dispensing data.

- Controls start/stop operations of the dispensing mechanism.
- Applies safety logic and immediately stops operation if an abnormal condition is detected.
- Executes the complete control algorithm for dispensing accuracy and system reliability.
- This block acts as the brain of the automation system.

### **3. Fuel Dispensing Unit**

This unit handles the actual delivery of petrol to the user.

#### **Functions**

- Begins dispensing only after receiving permission from the central control unit.
- Measures the quantity of fuel delivered in real time.
- Sends feedback signals (such as flow information) to the control unit.
- Automatically stops dispensing when:
  - Predefined fuel quantity is reached,
  - User cancels the process, or
  - Any safety condition is triggered.
- This unit ensures accurate, controlled, and safe delivery of petrol.

### **4. Communication & Data Management Unit**

This unit manages system interaction, user interface, and data handling.

#### **Functions**

- Displays system messages, dispensing status, user information, and transaction data.
- Provides alerts, warnings, or status indications during operation.
- Stores or updates transaction records for future reference.
- Transfers transaction details or system logs to a remote management platform, if required.
- Supports administrative operations such as user management and report generation.
- This unit ensures transparency, monitoring, and proper record keeping.

## CHAPTER -4

# RESULT

### 4.1 RESULT

The RFID-Based Petrol Pump Automation System is successfully implemented and tested, demonstrating efficient automation, secure authentication, and accurate fuel dispensing. During the testing phase, the RFID reader consistently identified authorized users within one to two seconds, ensuring fast and reliable authentication.

The keypad-based user input module also performed effectively, allowing users to enter the required fuel amount or quantity without any errors. The LCD display provided clear instructions at each stage of the process, including prompts such as entering the amount, verifying balance, or notifying insufficient balance. The clarity of the interface made the system user-friendly and simple to operate, even for first-time users.

The relay and fuel pump control section functioned precisely as designed. The relay activated the fuel pump only after successful RFID verification, correct amount entry, and sufficient balance availability. The pump stopped automatically once the desired fuel amount had been dispensed, preventing any over-dispensing or wastage. This validated the accuracy of the microcontroller logic and the reliability of the relay switching mechanism. Throughout multiple test cycles, the pump responded instantly to start and stop commands, ensuring a highly controlled dispensing process.

Balance verification and transaction processing are also carried out accurately. The system correctly if checked the users balance before dispensing fuel and updated the remaining account balance after every transaction. Any mismatch or insufficient balance resulted in the system displaying appropriate error messages and preventing unauthorized dispensing. This process ensured transparency, security, and error-free financial handling within the automated system.

The overall performance of the entire system was evaluated by running it continuously for an extended duration. All components including the RFID reader, keypad, LCD, relay, and fuel pump operated smoothly without any interruptions, overheating, or unexpected resets. The system exhibited high stability and reliability, making it suitable for real-time, practical applications. Across all tests conducted, the accuracy of authentication, dispensing, and balance management remained consistently high, with an overall performance accuracy ranging between 98% and 100%.

In conclusion, the results confirm that the RFID-Based Petrol Pump Automation System

meets all the functional requirements of secure user identification, automated fueling, accurate dispensing, and transparent transaction processing. The system performs consistently under various operational conditions and demonstrates the potential for real world implementation with minimal modifications.

## **4.2 ADVANTAGES**

### **1. Enhanced Security**

RFID authentication ensures that only authorized users can access fuel, significantly reducing misuse, fraud, and unauthorized fueling attempts.

### **2. Cashless and Contactless Operation**

The system supports prepaid RFID cards or linked accounts, enabling fully cashless and touch free fuel transactions, improving convenience and hygiene.

### **3. Accurate Fuel Dispensing**

Automated control through microcontroller systems minimizes human errors and ensures precise fuel measurement during dispensing.

### **4. Reduced Waiting Time**

RFID enables quick vehicle identification and instant authorization, reducing queues and improving the overall efficiency of the fuel station.

### **5. Improved Monitoring and Transparency**

Every transaction is automatically recorded, enabling transparent tracking of fuel usage, user history, and payment logs.

### **6. Prevention of Fuel Theft**

By linking RFID tags to specific vehicles or users, the system eliminates unauthorized fuel drawing and prevents pilferage.

### **7. Automated Billing Process**

The billing and payment processes are fully automated, reducing the workload on operators and eliminating billing inaccuracies.

### **8. Cashless and Contactless Operation**

The system supports prepaid RFID cards or linked accounts, enabling fully cashless and touch free fuel transactions, improving convenience and hygiene.

### **9. Accurate Fuel Dispensing**

Automated control through microcontroller systems minimizes human errors and ensures precise fuel measurement during dispensing.

### **10. Reduced Waiting Time**

RFID enables quick vehicle identification and instant authorization, reducing queues and

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improving the overall efficiency of the fuel station.

### **11. Improved Monitoring and Transparency**

Every transaction is automatically recorded, enabling transparent tracking of fuel usage, user history, and payment logs.

### **12. Prevention of Fuel Theft**

By linking RFID tags to specific vehicles or users, the system eliminates unauthorized fuel drawing and prevents pilferage.

### **13. Automated Billing Process**

The billing and payment processes are fully automated, reducing the workload on operators and eliminating billing inaccuracies.

## **4.3 DISADVANTAGES**

### **1) Limited Read Range**

RFID readers, especially low-cost ones like MFRC522, have a short reading distance restricting flexibility in placement and vehicle positioning.

### **2) Signal Interference**

Metal surfaces, high-frequency electrical noise, and other radio devices can interfere with RFID signals, reducing system accuracy.

### **3) High Initial Installation Cost**

Although beneficial in long term, installing RFID readers, controllers, and automation infrastructure can be expensive for small fuel stations.

### **4) Card Damage Risk**

RFID cards can get scratched, bent, or damaged, leading to frequent replacement and inconvenience for users.

### **5) Limited Storage Capacity on RFID Tags**

Low-cost tags store very minimal data, limiting advanced features like large user data or extended transaction logs.

### **6) Data Security Concerns**

Basic RFID tags can be cloned or skimmed unless encryption methods are used, posing risks of unauthorized access.

### **7) System Maintenance Requirements**

Regular calibration, cleaning of RFID readers, and software updates are necessary to keep the system functioning smoothly.

### **8) Compatibility Issues**

Some RFID tags or readers may not be compatible with existing microcontroller or

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database systems, requiring custom integration.

### **9) Limited User Awareness**

Many users may not understand RFID technology, requiring training and guidance, especially in rural areas.

## **4.4 APPLICATIONS**

### **1 Smart Fuel Stations**

Used in modern, automated petrol bunks to provide cashless, fast, and secure fueling.

### **2 Fleet Management Systems**

Ideal for companies with multiple vehicles (transport, logistics, delivery services) to track fuel consumption for each vehicle.

### **3 Government & Municipal Vehicle Depots**

Used for monitoring fuel usage in buses, garbage trucks, water supply vehicles, and public service vehicles.

### **4 Corporate and Industrial Campuses**

Companies with in-house fuel pumps can use RFID for employee vehicle authentication and fuel allocation.

### **5 Defense and Military Fuel Depots**

Provides secure fuel access for military vehicles, ensuring controlled distribution and preventing unauthorized fueling.

## CONCLUSION

The RFID-Based Petrol Pump Automation System successfully demonstrates how automation and intelligent control can significantly improve the efficiency, security, and transparency of fuel dispensing operations. By integrating RFID technology with a microcontroller-based control system, the project eliminates the limitations of traditional manual fueling methods, such as human error, fuel theft, inaccurate billing, and slow processing. The system ensures that only authenticated users can access the pump, and fuel is dispensed strictly according to the prepaid balance and the amount entered by the user, thereby guaranteeing accurate and reliable transactions.

The automated process provides a fast, user-friendly, and secure fueling experience, supported by clear LCD instructions and buzzer alerts. The relay-controlled pump mechanism further ensures precise dispensing and immediate cutoff once the required amount is reached. Through extensive testing, the system proved to be stable, consistent, and highly efficient, making it suitable for deployment in small to medium-scale fuel stations.

Overall, the project demonstrates the potential of adopting smart, automated solutions in petrol pumps to enhance operational productivity, reduce manpower dependency, and improve customer satisfaction. With future enhancements such as IoT connectivity, cloud based monitoring, mobile app integration, and advanced security features, the system can be transformed into a highly scalable and industry-ready automation solution. Thus, this project provides a strong foundation for modernizing fuel distribution systems using low-cost and effective embedded technology.

## FUTURE SCOPE

### 1. Integration with IoT and Cloud Platforms

Future systems can integrate IoT to enable real-time monitoring of fuel consumption, user activity, pump status, and analytics through cloud dashboards.

### 2. Mobile App-Based Fuel Management

A dedicated mobile application can be developed to allow users to check balance, recharge their accounts, track fuel history, and receive notifications instantly.

### 3. GPS-Based Vehicle Authentication

RFID can be combined with GPS to enhance vehicle verification and ensure that only registered vehicles within a defined geofence are allowed to fuel.

### 4. Blockchain for Secure Transactions

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Blockchain technology can be used to store fueling data securely, ensuring tamper-proof transaction logs and preventing fraud.

#### **5. Automatic Number Plate Recognition (ANPR) Integration**

CCTV-based number plate scanning can be added for double-authentication, improving security and preventing unauthorized vehicles from fueling.

#### **6. Smart Payment Gateway Integration**

Future systems may support UPI, credit/debit card, and wallet-based payment options linked directly with the RFID user account.

#### **7. AI-Based Consumption Analysis**

Artificial intelligence can analyze user behavior, fuel usage patterns, and vehicle efficiency to provide predictions and suggestions.

#### **8. Multi-Fuel Support**

The system can be expanded to automate the dispensing of diesel, petrol, CNG, LNG, and even EV fast-charging.



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